

Thread - 2026-03-05

<https://x.com/RiikonenMarko/status/2029581786833953249>

1/8 - 2026-03-05 15:36 UTC

4/5 March 2026, Rovaniemi. As the evening progressed to night, the cloud layer lowered. Woke up at 0230 and checked FMI graphs. Airport had changed from 8/8 to 9/8, but I could have just as well looked out of the window: the radio-mast top was enshrouded and Suosiola plume mixed with the cloud.

Time to go, then. Temp between -4 and -5 C, so from the previous nights experiences I expected whatever action there might be, be tied to the very proximity of snow gun.

That was the case. And this time the display never took off. Just tanarc, didn't see any diffuse arc, the opposite direction was uncannily glitterless even during the best tanarc boosts. If crystals are optically good quality, you see diffuse arc accompany even weakest tanarc in sparse crystals.

The action was either

- 1) just fog
- 2) short tanarc boosts
- 3) snow-fall like boosts of matte crystals
- 4) or sparse glitter in big crystals that sketched poorly a wide pillar with some suggestion of tanarc.

That's only rough characterization, of course. These stages were also mixed.

The three images show 2-4. The stage 2 image is single frame, stage 3 is 6 frame ave, and stage 4 is 16 frame max. I took a crystal sample because I was curious about two things: why no diffuse arc & whatever other basal face reflection requiring halos; and what kind were those big crystals. I'll continue the thread later with crystal photos.

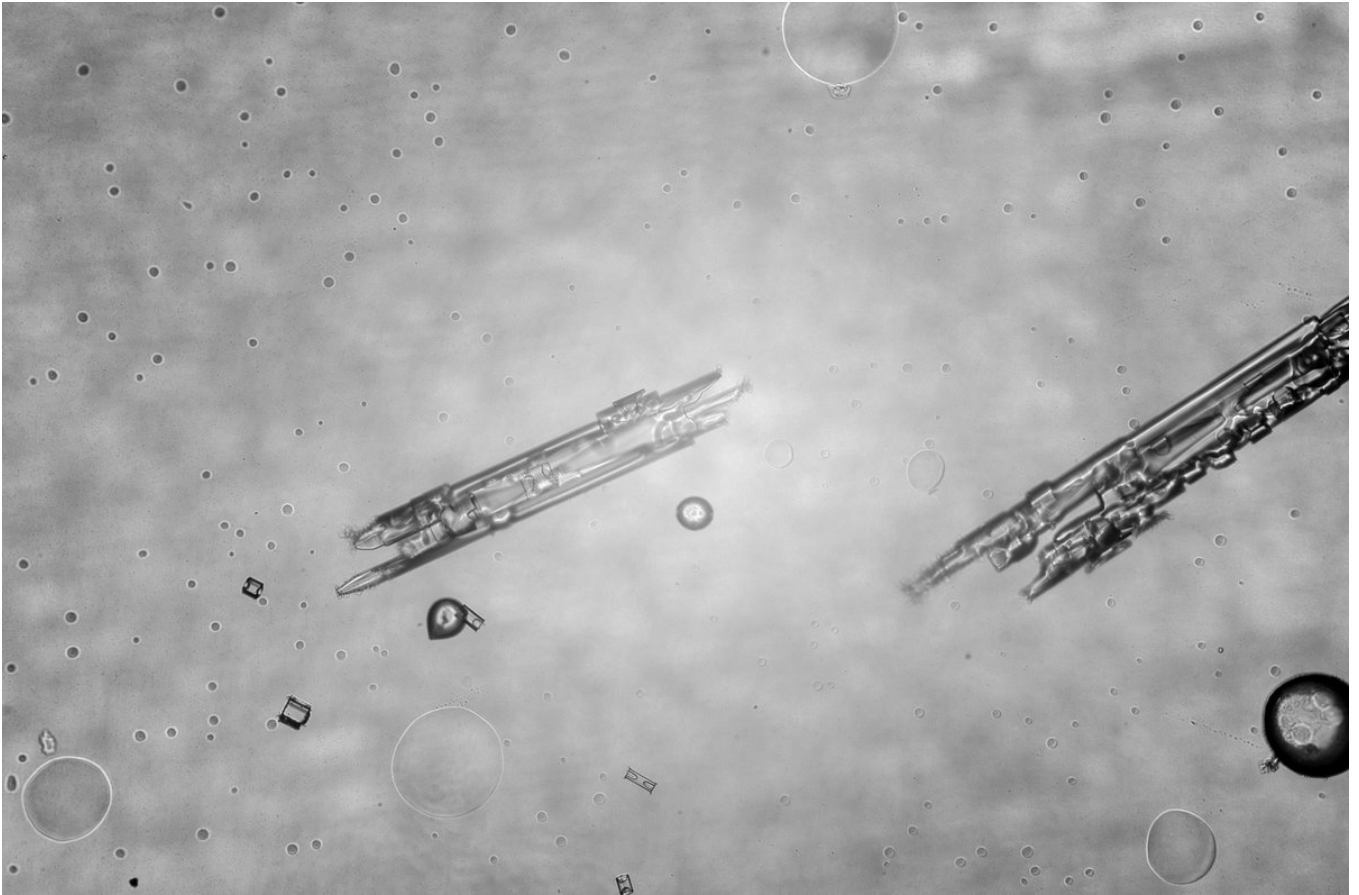
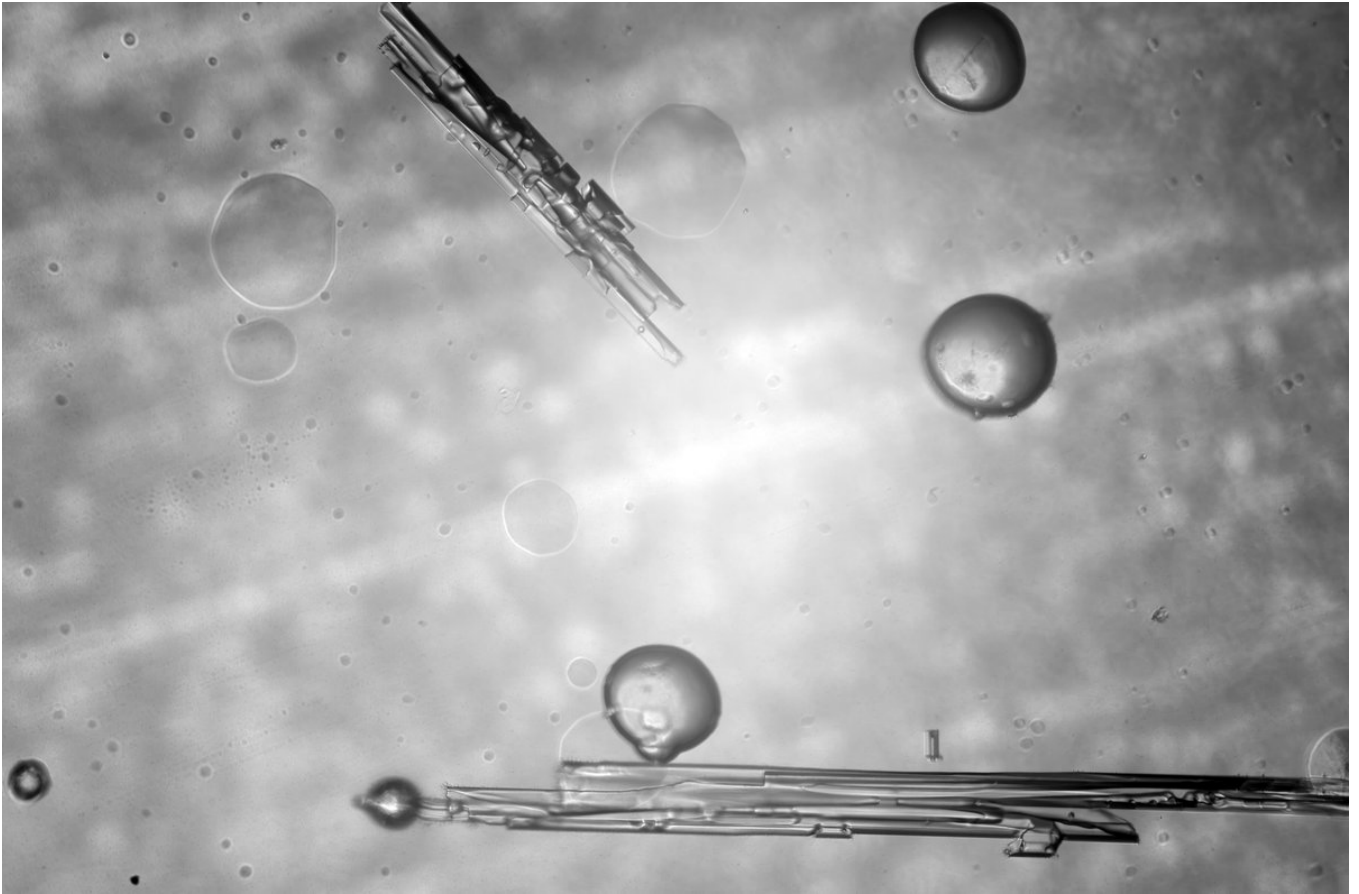


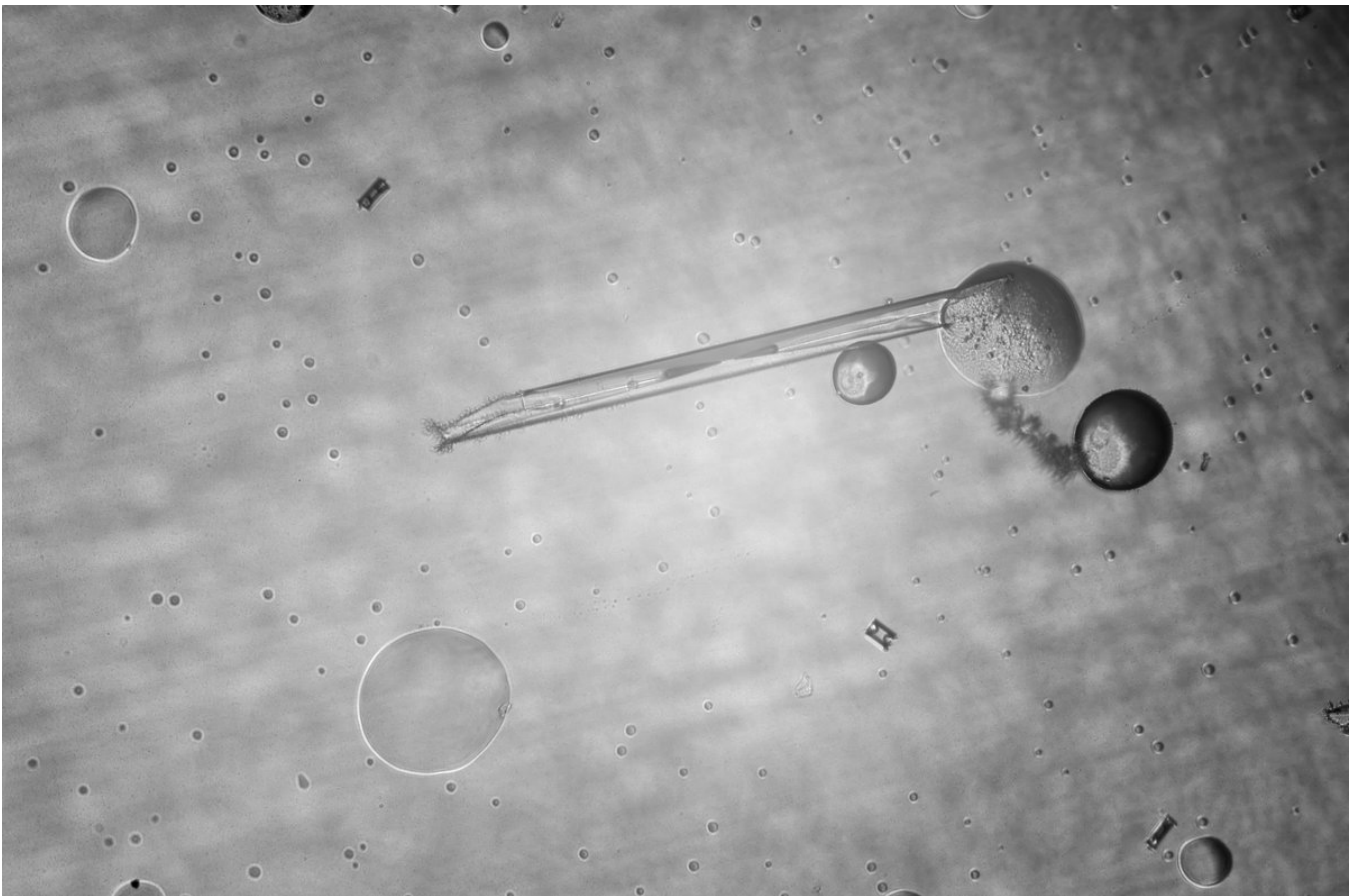




2/8 - 2026-03-05 16:13 UTC

4/5 March 2026, Rovaniemi. An so on to the crystals. To be true, I knew already before looking through the microscope what kind of crystals made the stage 4 display because I could see there were long columns on my jacket sleeve. Yeah, the only crystals in the sample that fit the bill for stage 4 are those gigantic imperfect columns.

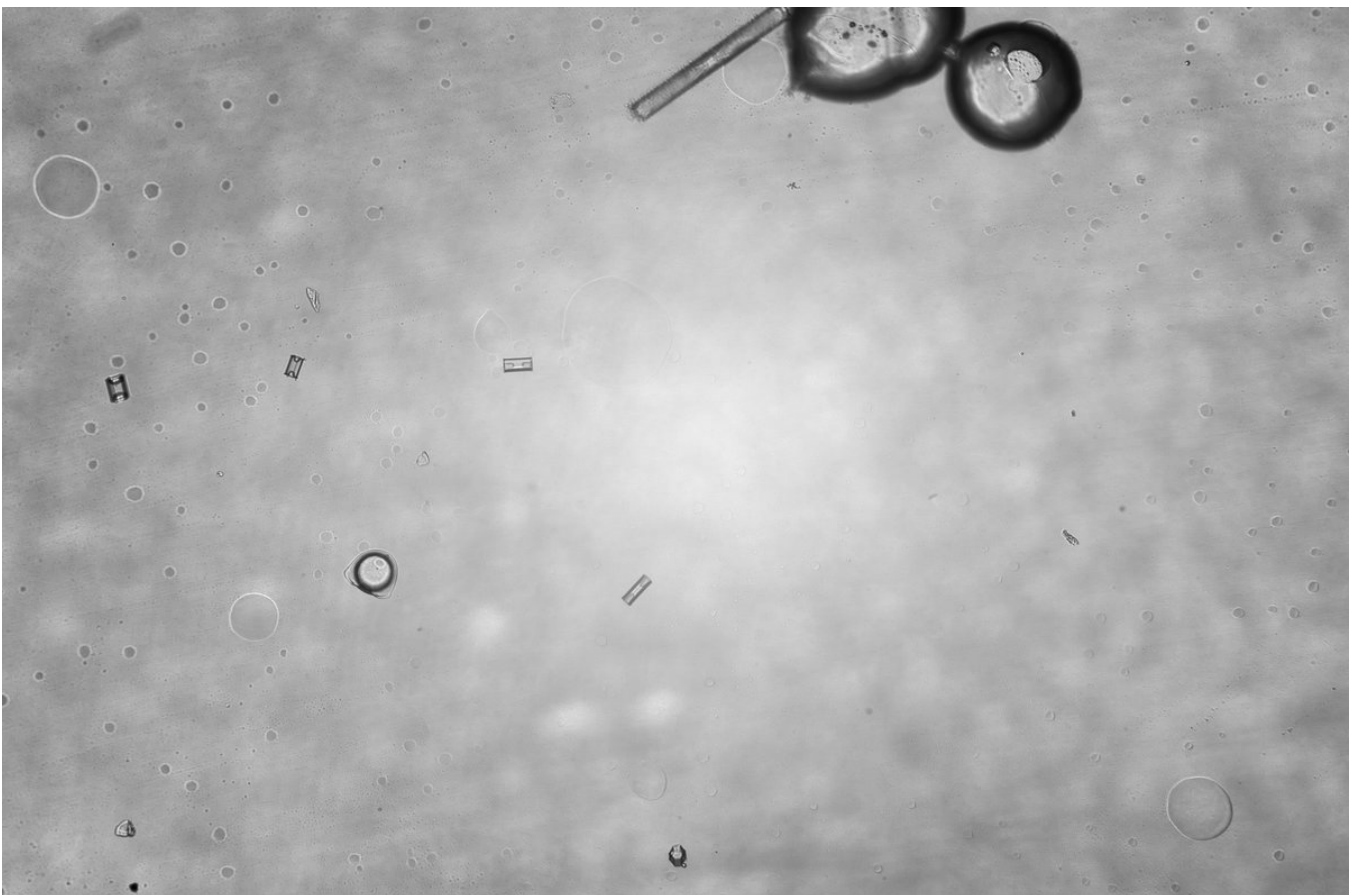
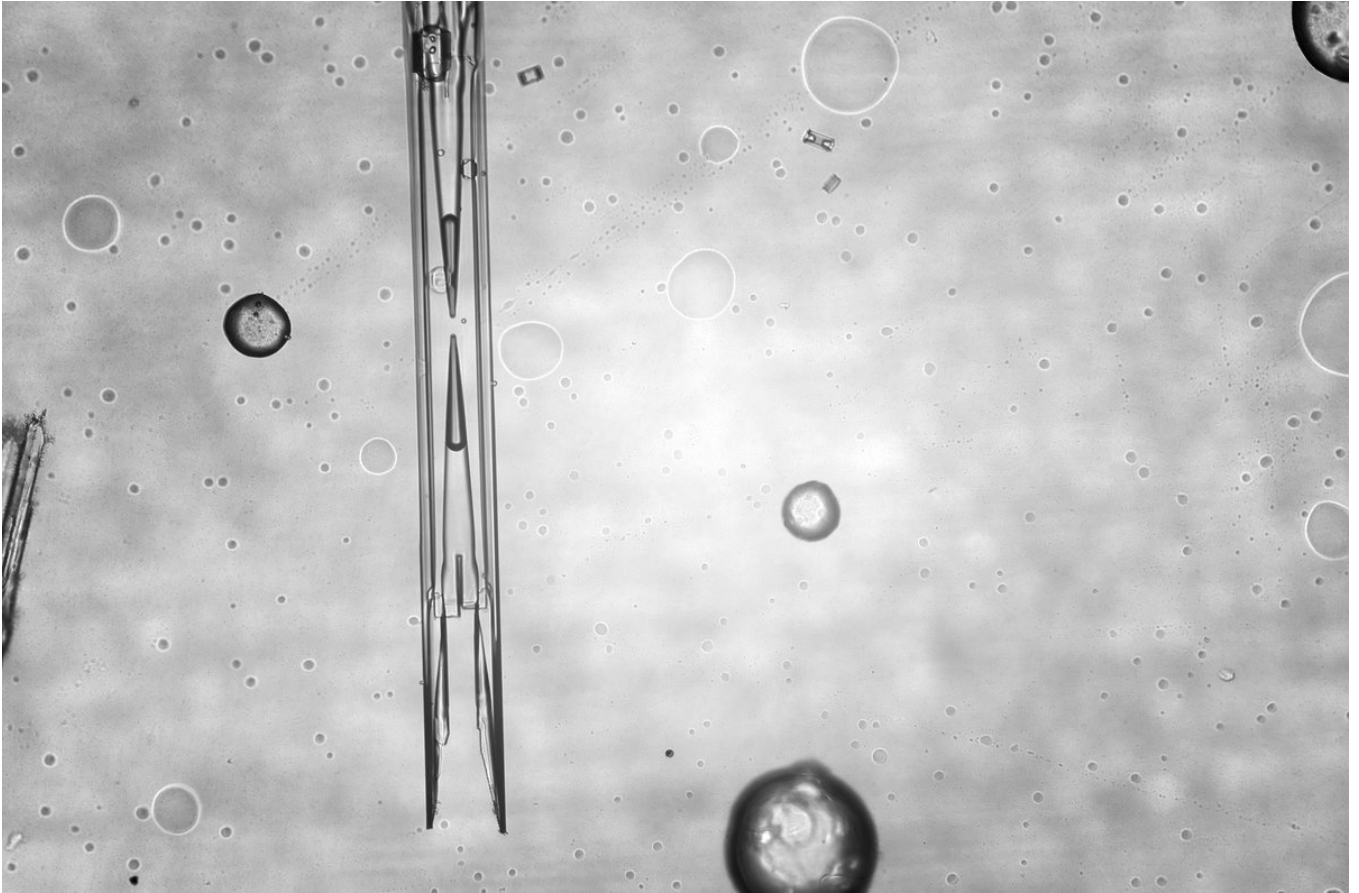


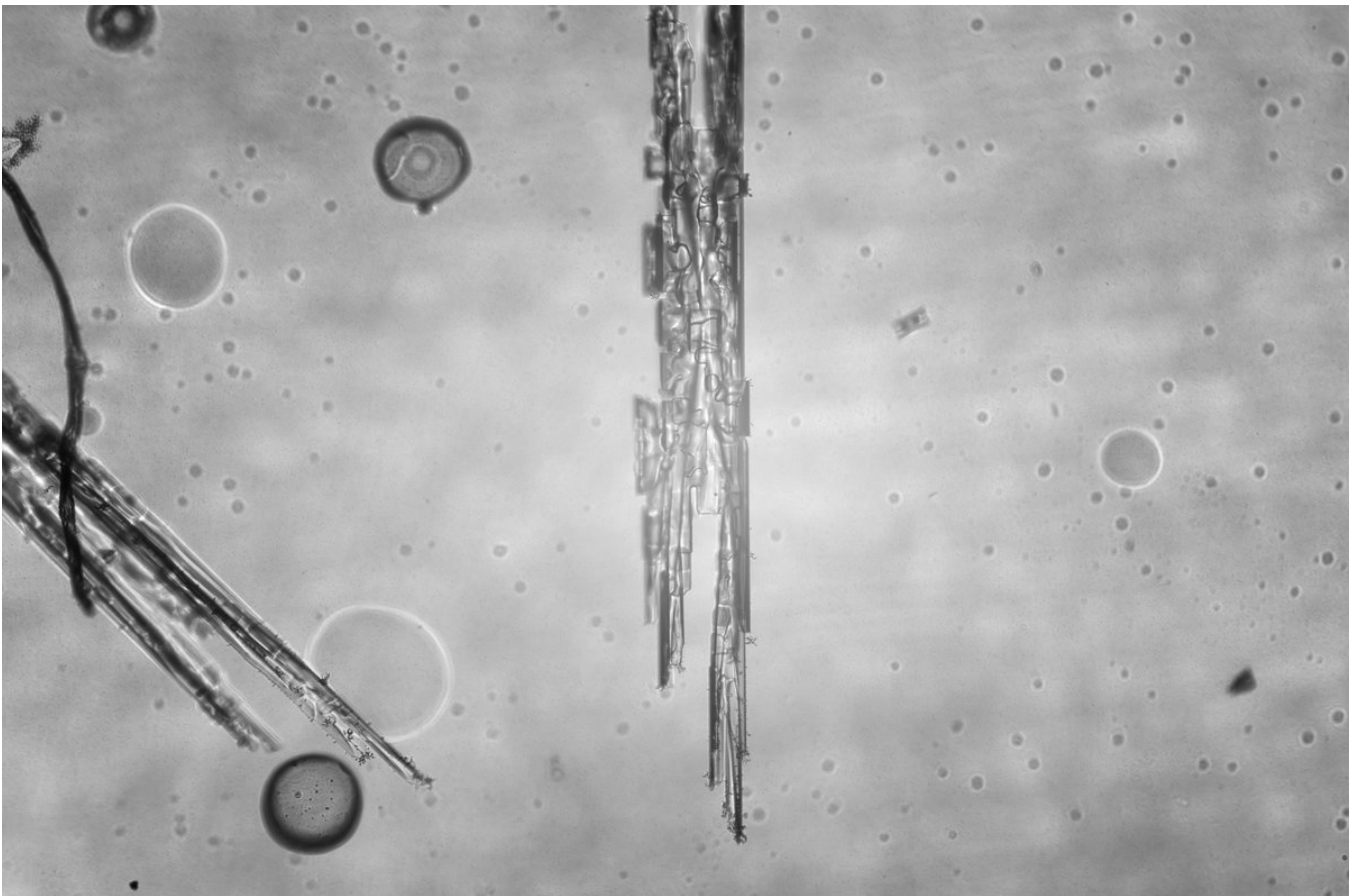
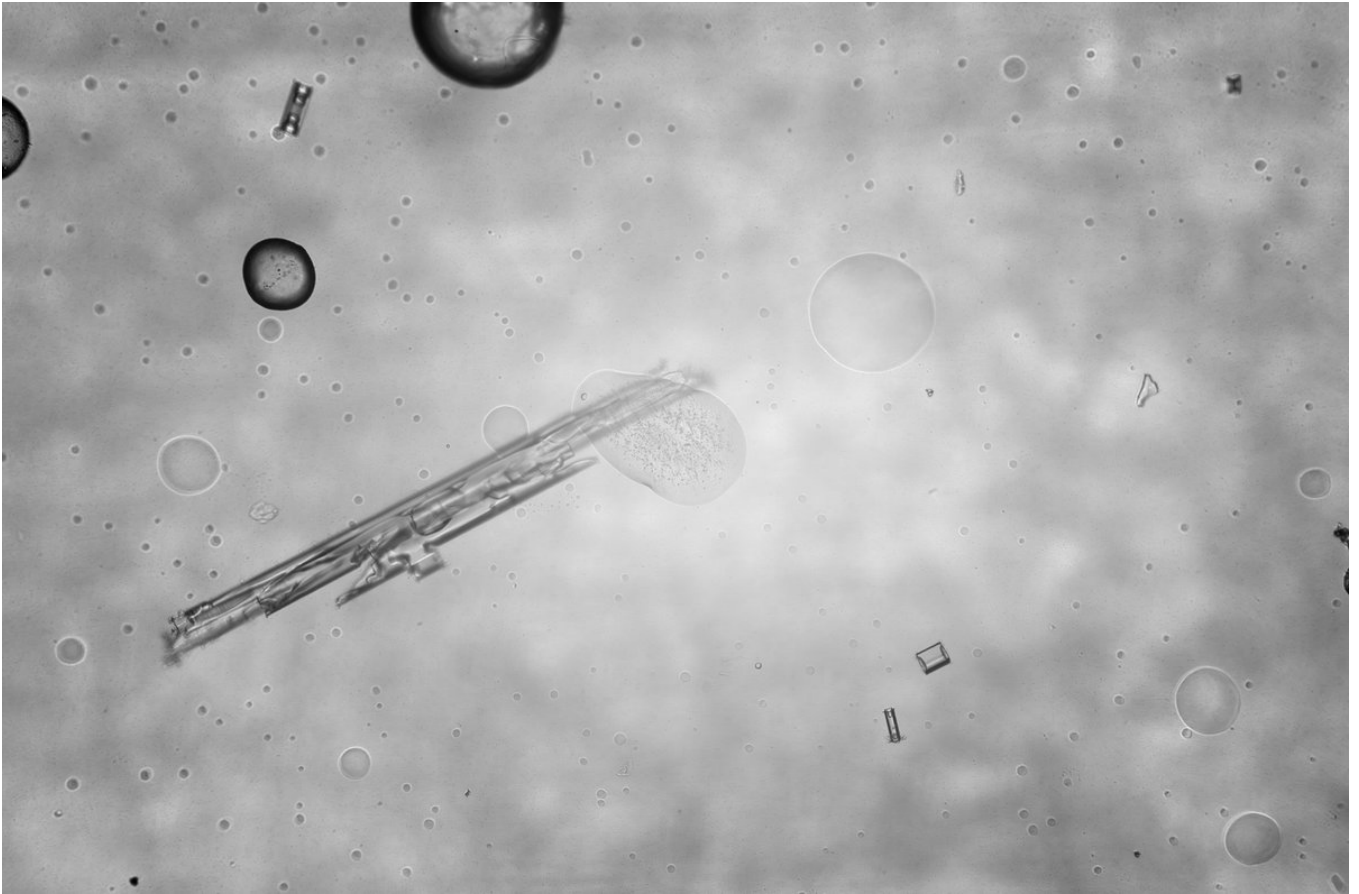


3/8 - 2026-03-05 16:21 UTC

4/5 March 2026, Rovaniemi. The tanarc, in turn, must have been made by the small columns. And the

end cavities meant that diffuse arc and all other all basal face reflection requiring halos, such as any 46° stuff, were absent. It may be possible that end cavities do not result in an absence of basal face reflection requiring halos if the cavities are stepped, but here they are rounded.





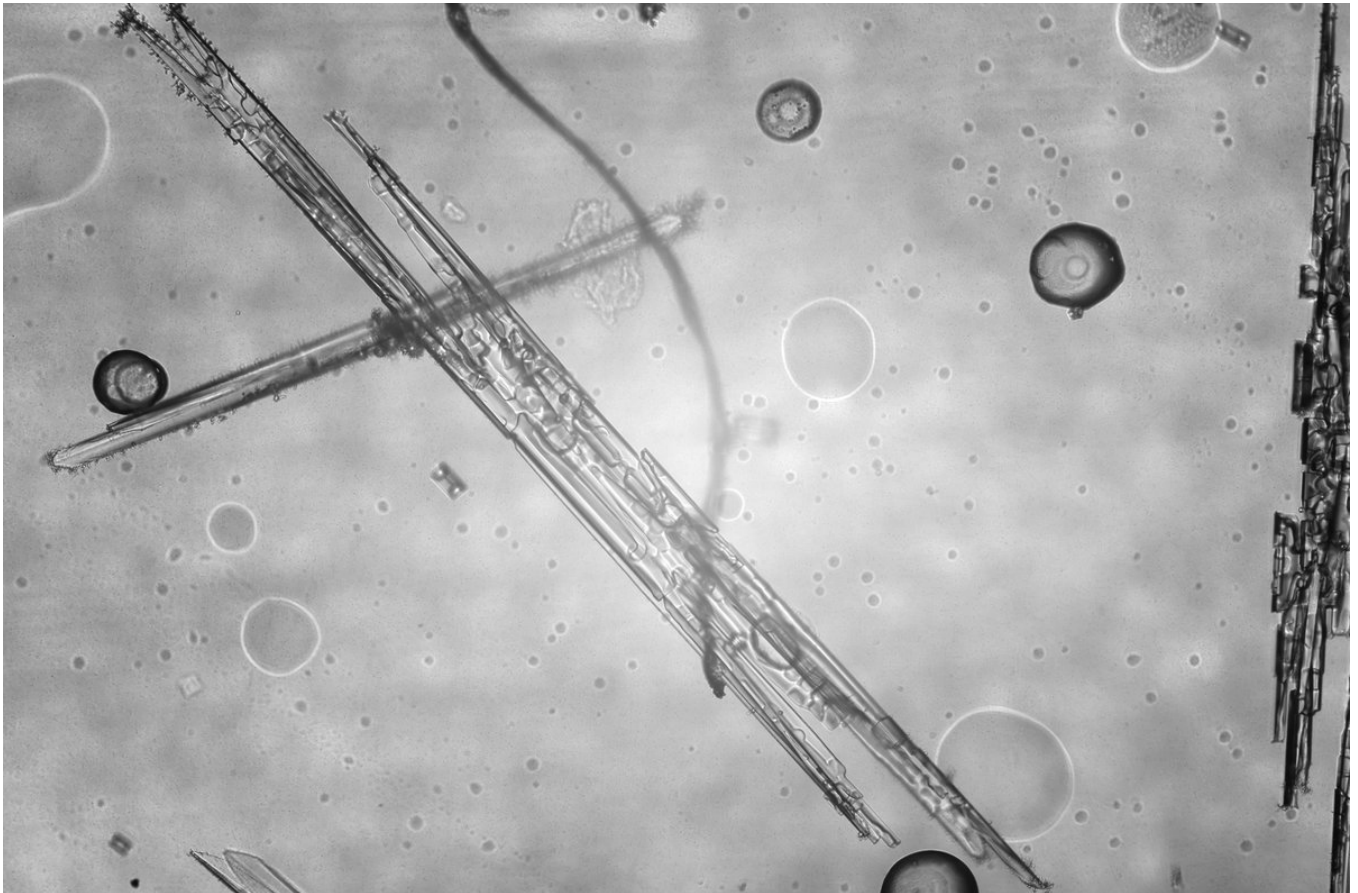
4/8 - 2026-03-05 17:10 UTC

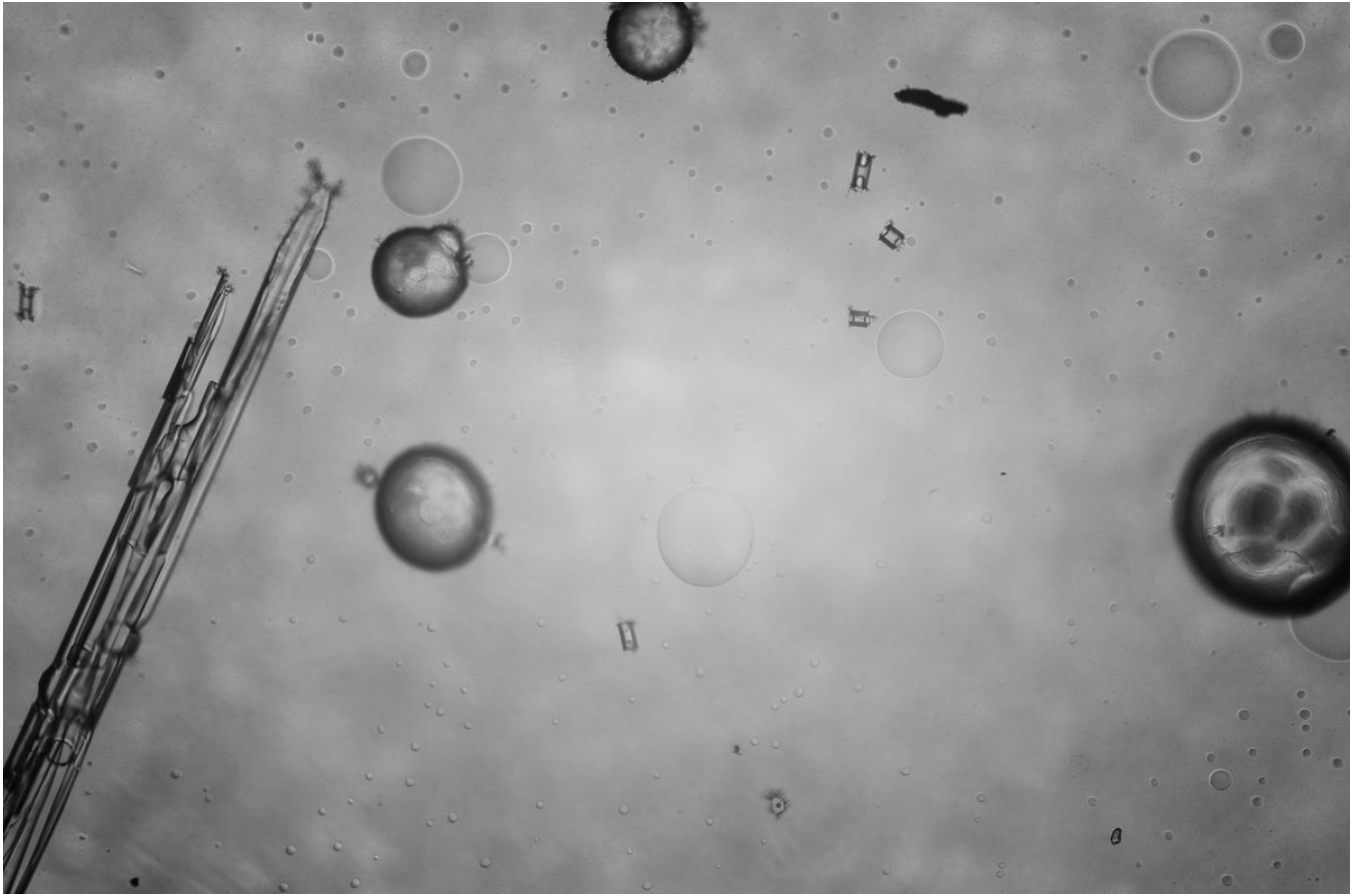
4/5 March 2026, Rovaniemi. As to the snowfall like bursts, it must have been those frozen droplets. I

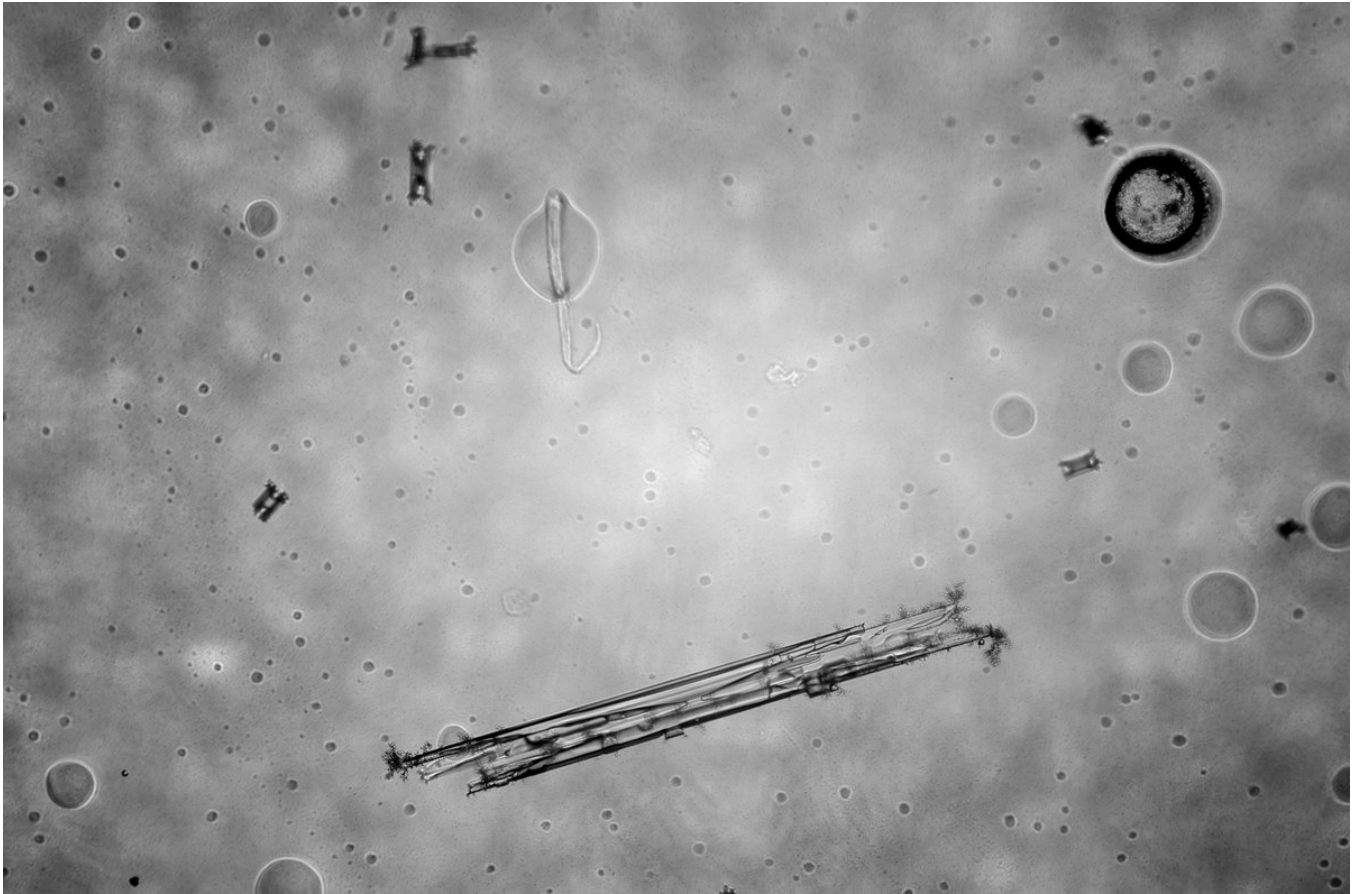
am not sure if this stuff is straight from the snow gun nozzle or if natural fog droplets were frozen. In case of the latter there must have been significant occasional increases in fog droplet size.

One might wonder whether frozen droplets would be able to make a mimic fogbow. That would be an exciting new halo. If the stage 3 matte bursts were indeed made by these droxtals, then no – I looked and didn't see no fogbow opposite to the lamp.

But many of those droxtals look uncomfortably round. You ask AI and it tells "Yes, droxtals can definitely make fogbows". But there doesn't seem to be any actual evidence. Do we have here an issue that needs some experimental studies?



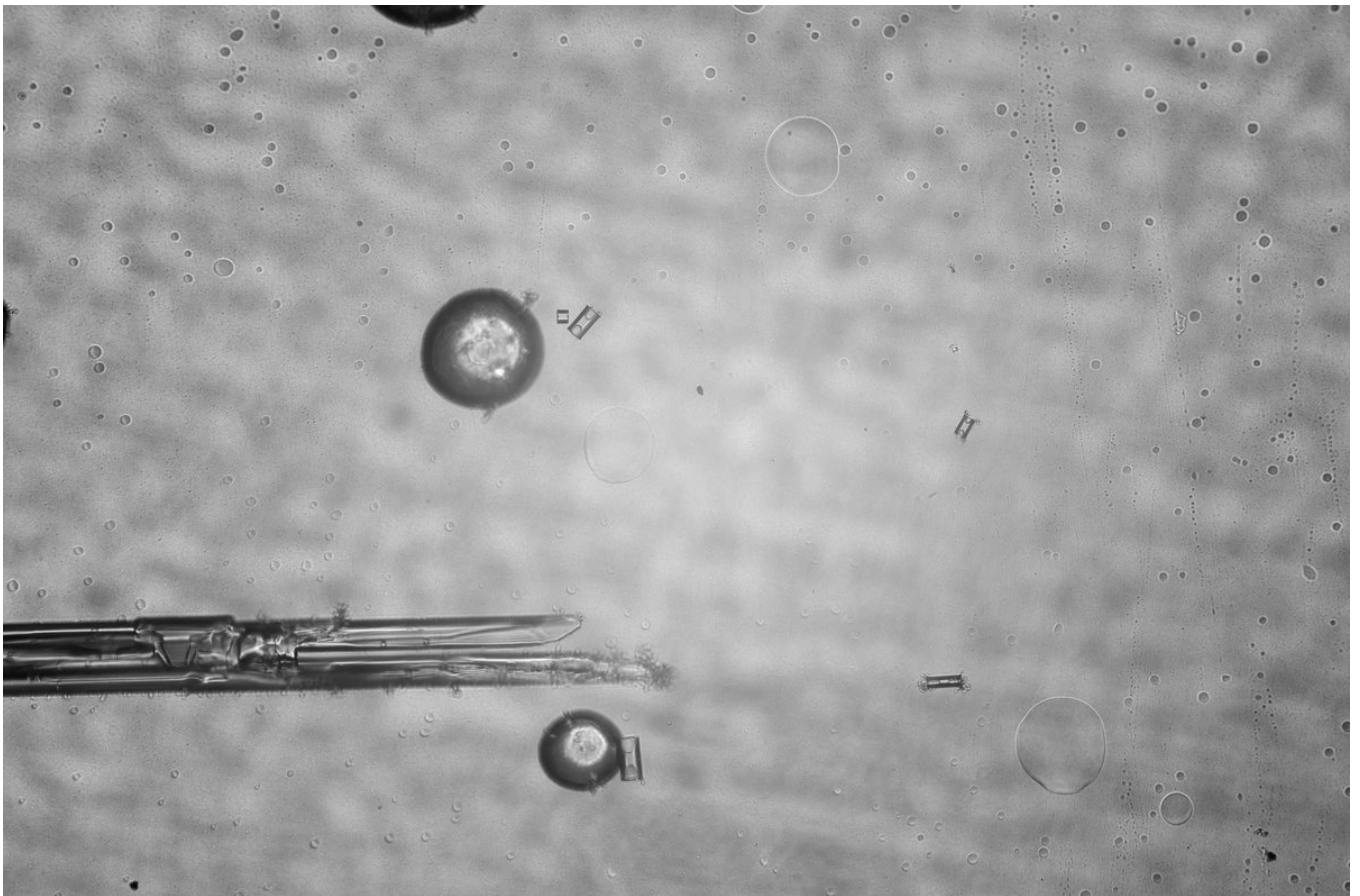
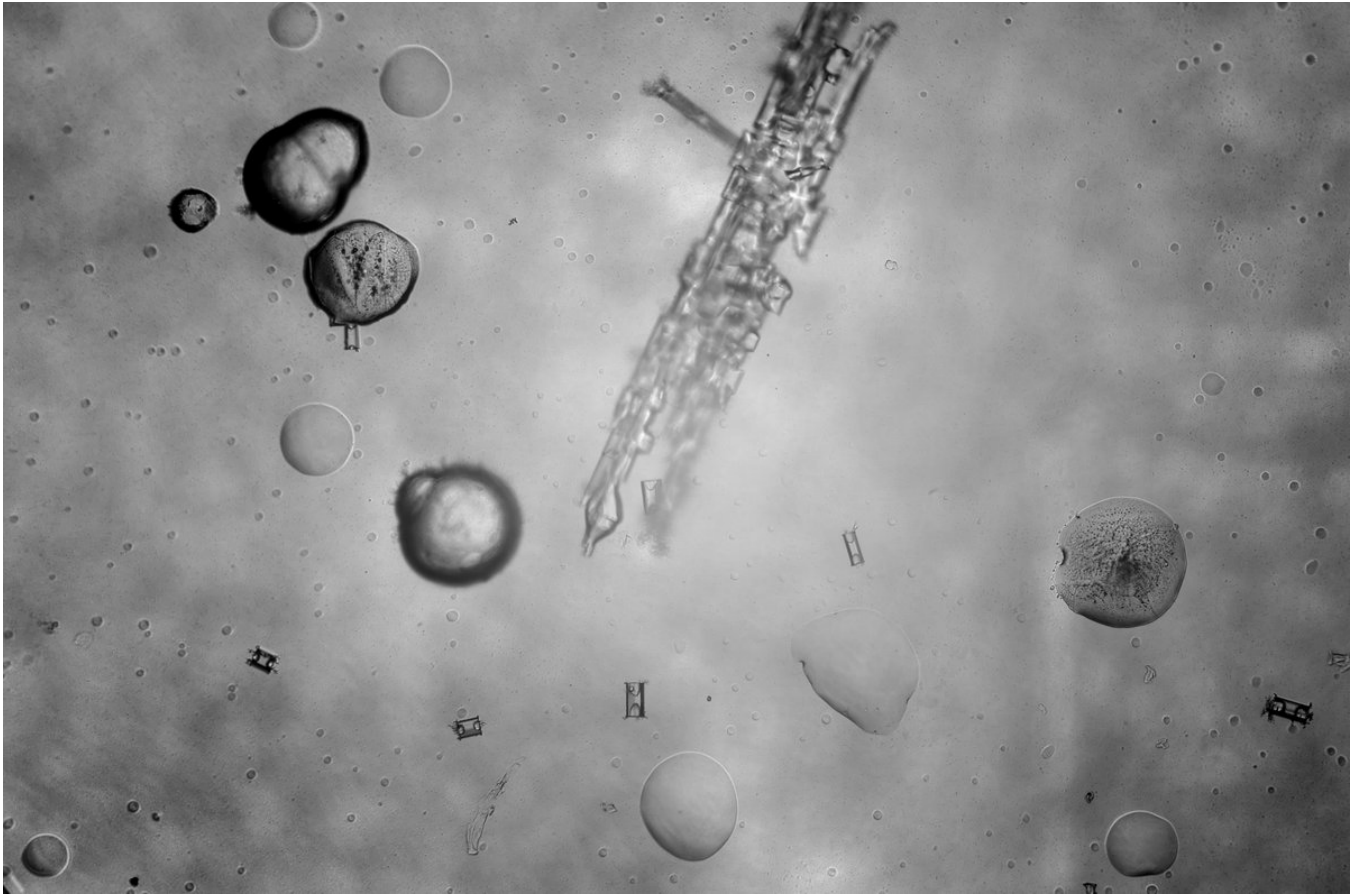


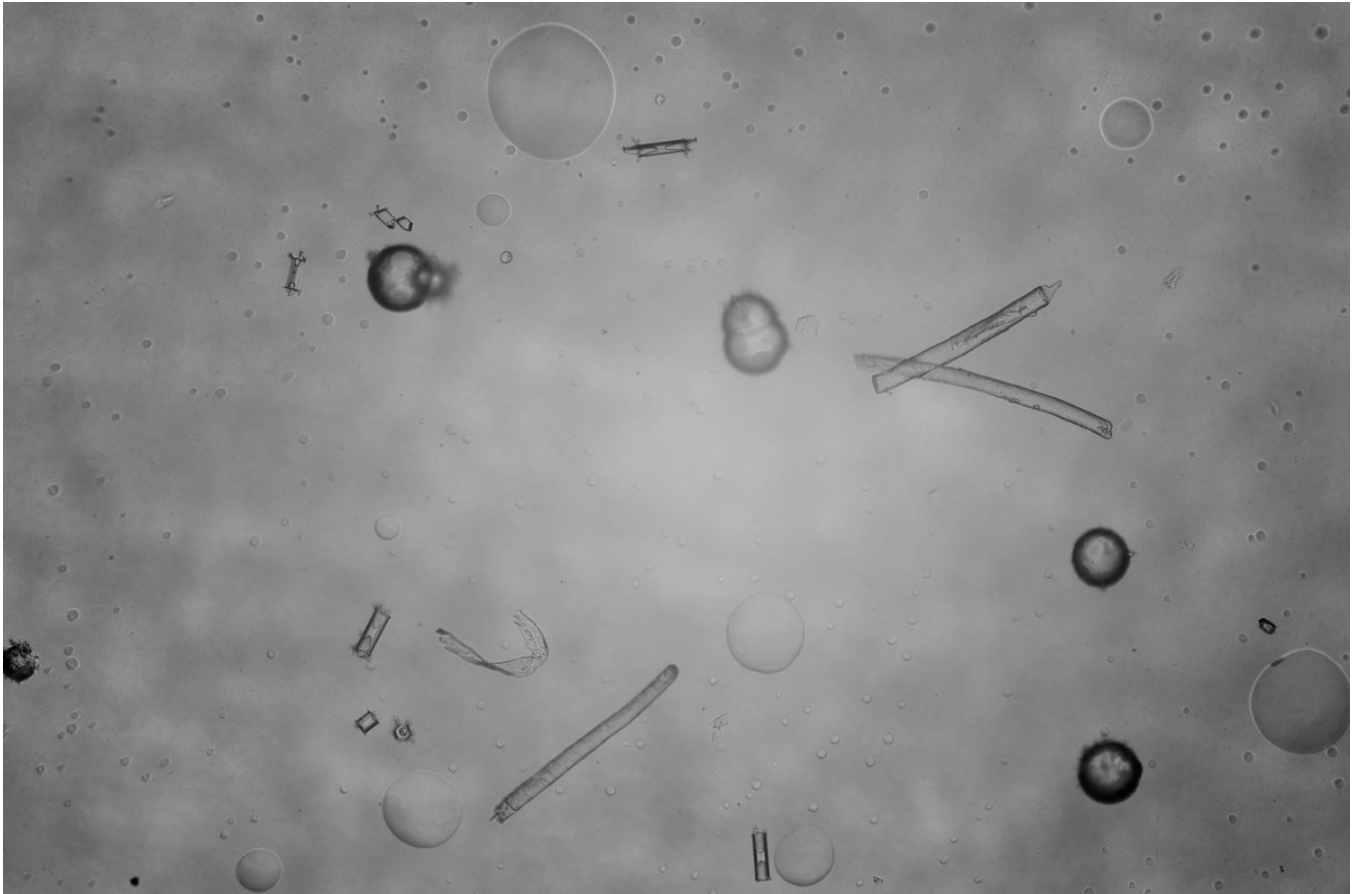


5/8 - 2026-03-05 17:11 UTC

4/5 March 2026, Rovaniemi. More crystals.

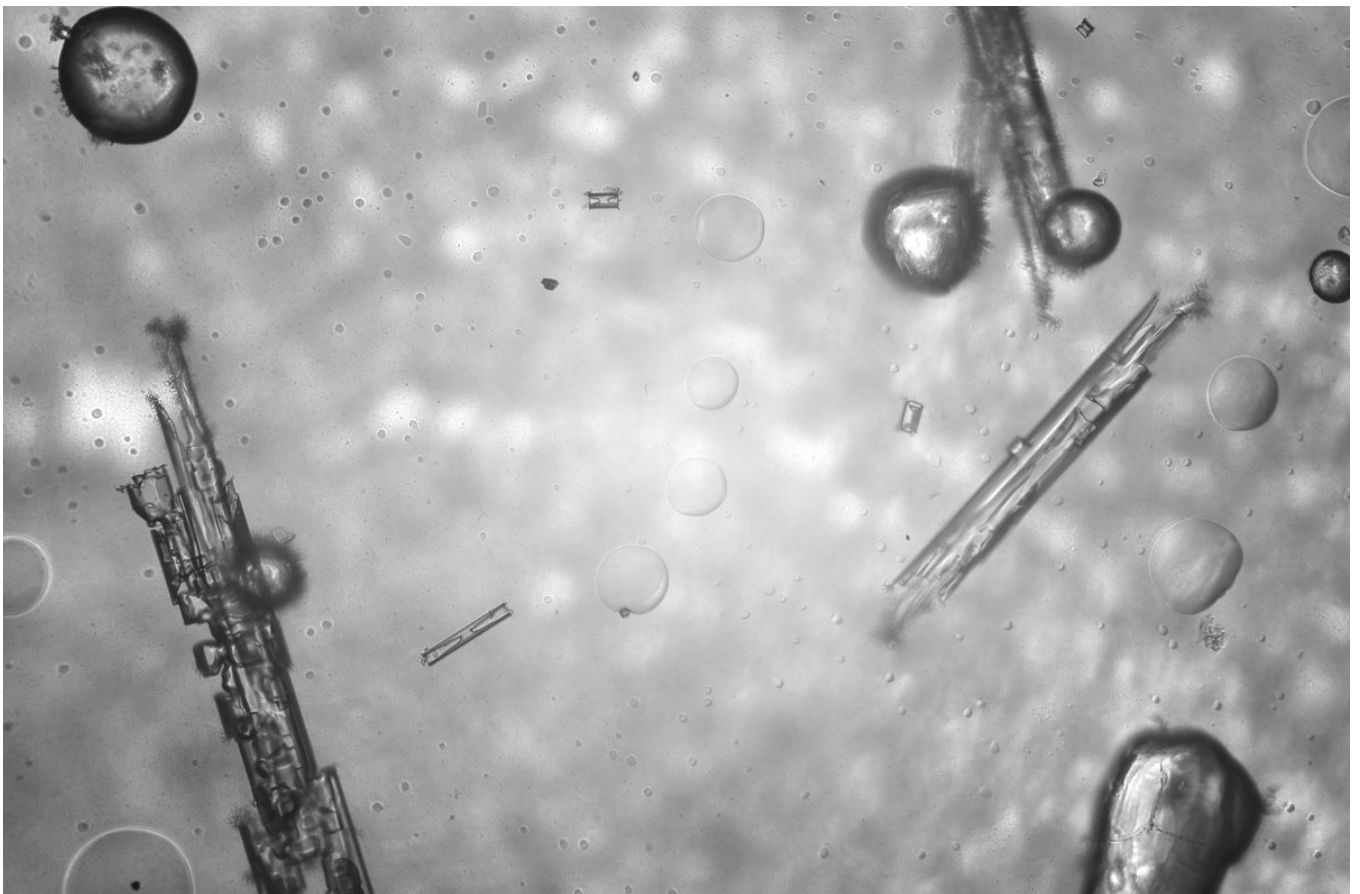






6/8 - 2026-03-05 17:12 UTC

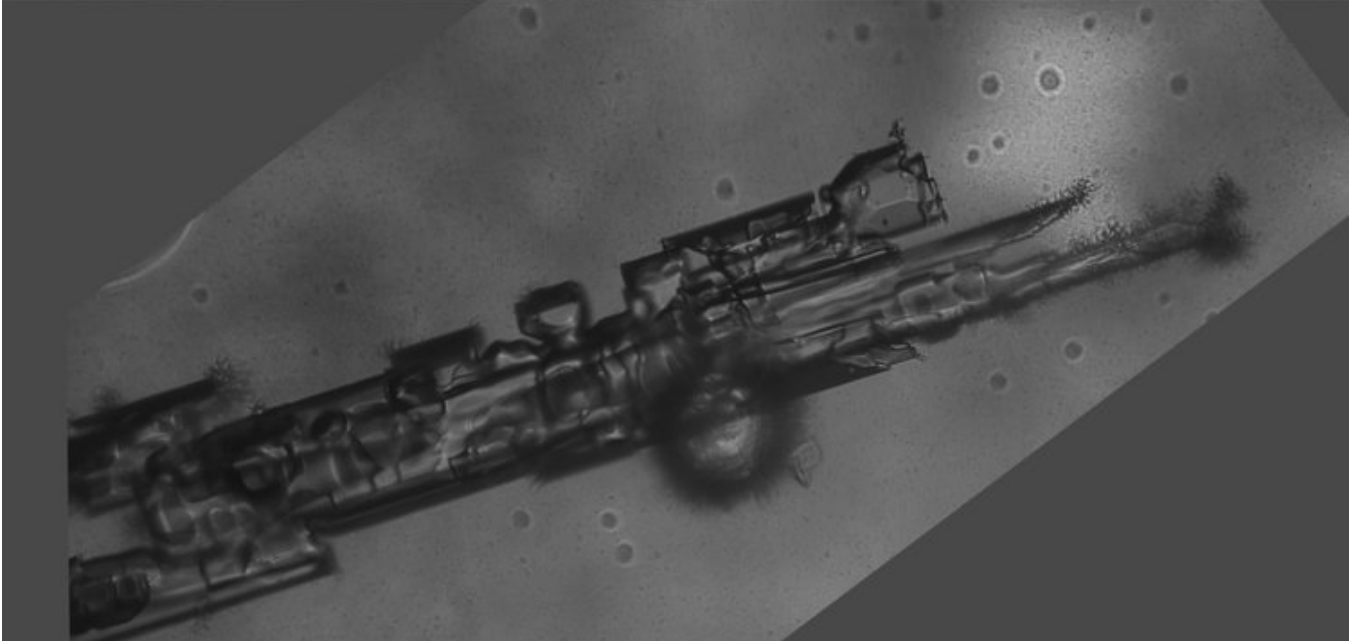
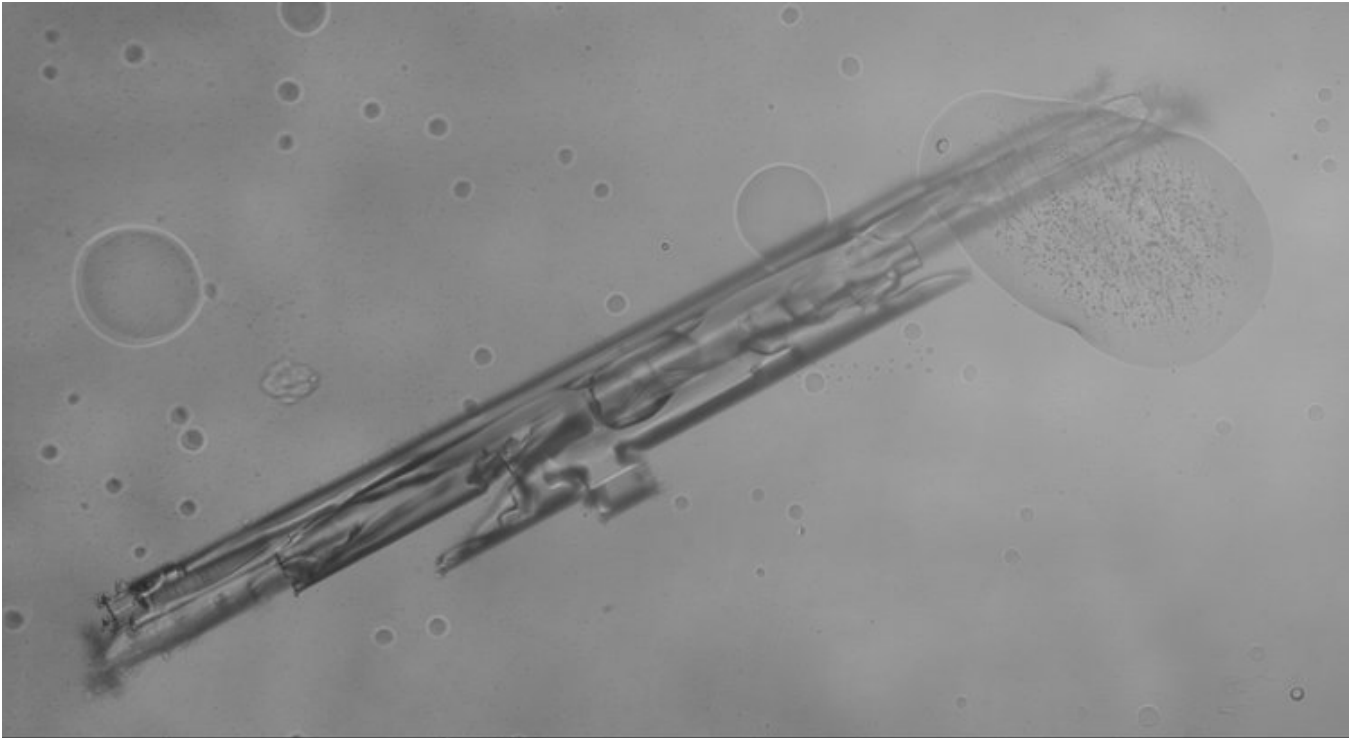
4/5 March 2026, Rovaniemi. Last of the crystals.





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4/5 March 2026, Rovaniemi. Sulaco!



8/8 - 2026-03-06 11:10 UTC

4/5 March 2026, Rovaniemi. All frames from the crystal sampling period (which includes the time I photographed the crystals in the open). Green dot marks the point at which I started crystal photography. The second row from the bottom of the first collage presents a longer stage when the display was made of only gigantic columns. Something grotesquely gothic about those crystals, really.

In all, I consider this among the top interesting crystal samples I have taken. Well defined tanarcs without any basal face reflection halos are highly rare in diamond dust and it is a great opportunity if you can have a look at the crystals to see what's the reason. And of course it was satisfying to be able photograph also the big ones and be able have an unalloyed display stage from them.

